

EC341 · SVNIT SURAT · THE TU CHAHIYE QUIZ, ON PAPER

DIGITAL IMAGE PROCESSING

40 QUESTIONS

90 MARKS

45 MINUTES

ANSWER KEY AT THE BACK

How the marks work. Ten easy questions at 1 mark, ten medium at 2, and twenty hard at 3 — ninety in total, two thirds of them in the hard twenty. One option is right in every question. **The answer key, with a short explanation for every question, starts on the last section of this document.**

40

QUESTIONS

90

MARKS

45:00

ON THE CLOCK

1

ATTEMPT ONLINE

- 01** In the image model $f(x, y)$, what does the value of f at a point represent? 1 MARK
- A The position of that pixel in the row B The intensity, or brightness, at that point
C The number of colour channels D The size of the image in bytes
- 02** In an 8-bit grayscale image, which value is pure black? 1 MARK
- A 255 B 128
C 0 D 1
- 03** Sampling and quantization digitise two different things. Sampling digitises the... 1 MARK
- A coordinates B amplitude
C file format D colour model
- 04** An image uses k bits per pixel. How many gray levels does it have? 1 MARK
- A k B $2k$
C 2 raised to the power k D k squared
- 05** How many pixels are in the set $N_4(p)$? 1 MARK
- A 2 B 4
C 8 D 9
- 06** Which distance measure is known as the city-block distance? 1 MARK
- A The Euclidean distance B D_4
C D_8 D D_{∞}

- 07** What is the negative transformation of an image with gray levels in $[0, L-1]$? 1 MARK
- A $s = L - 1 - r$ B $s = c \log(1 + r)$
 C $s = c r^\gamma$ D $s = r / (L - 1)$
- 08** Image enhancement is described as subjective. Why? 1 MARK
- A Because it uses random numbers B Because the viewer is the ultimate judge of how well a particular method worked
 C Because it can only be done by hand D Because it always destroys information
- 09** Which band of the electromagnetic spectrum has the highest energy per photon? 1 MARK
- A Radio waves B Microwaves
 C X-rays D Γ rays
- 10** In the histogram $h(r_k) = nk$, what is nk ? 1 MARK
- A The k th gray level itself B The number of pixels in the image having gray level r_k
 C The probability of gray level r_k D The total number of pixels in the image
- 11** How many bits are needed to store a 512×512 image with 256 gray levels? 2 MARKS
- A 262,144 B 524,288
 C 2,097,152 D 16,777,216
- 12** For $p = (2, 1)$ and $q = (6, 4)$, what are D_4 and D_8 ? 2 MARKS
- A 5 and 5 B 7 and 4
 C 4 and 7 D 5 and 4
- 13** A 3×3 matrix is convolved with a 2×2 kernel. What is the size of the result? 2 MARKS
- A 3×3 B 2×2
 C 4×4 D 6×6
- 14** An image is too dark. Which value of γ should be used in $s = c r^\gamma$? 2 MARKS
- A γ greater than 1 B γ less than 1
 C γ exactly 1 D γ equal to 0
- 15** Why is the histogram usually normalized by dividing every count by the total number of pixels? 2 MARKS
- A To make the image brighter B So that each value estimates the probability of that gray level, and the whole histogram sums to 1
 C To remove noise from the image D To convert the image to binary

- 16** An image is reduced from 1024×1024 to 256×256 . What percentage of the pixels is discarded? 2 MARKS

A 75 % B 93.75 %
 C 87.5 % D 99.6 %

17 In bit-plane slicing of an 8-bit image, which plane carries most of the visually significant information? 2 MARKS

A Plane 1, the least significant bit B Plane 4
 C Plane 8, the most significant bit D All planes carry equal information

18 What problem does m-adjacency solve? 2 MARKS

A It makes the image file smaller B It removes the ambiguity of the multiple paths that 8-adjacency allows
 C It speeds up convolution D It converts colour images to gray

19 The log transformation $s = c \log(1 + r)$ is mainly used to... 2 MARKS

A sharpen edges B compress the dynamic range, for example of a Fourier spectrum
 C convert an image to binary D rotate an image

20 A grayscale image is converted into an RGB image. What is the result? 2 MARKS

A The picture gains colour B The array becomes three-channel but all three channels are equal, so it still looks gray
 C The picture becomes brighter D The conversion is impossible

21 For the image segment below with $V = \{1\}$, are the pixels at (0,2) and (1,1) m-adjacent? Rows and columns are counted from 0. 3 MARKS

0 1 1
 0 1 0
 0 0 1

A Yes, because they are diagonal neighbours and both values are in V B No, because their common 4-neighbour (0,1) has the value 1, which is in V
 C Yes, because every 8-adjacent pair is also m-adjacent D No, because diagonal pixels can never be m-adjacent

22 Using the same segment and $V = \{1\}$, which statement about the route from (0,2) to (2,2) is correct? 3 MARKS

0 1 1
 0 1 0
 0 0 1

A There is a 4-path of length 2 B The shortest 8-path is 2 steps long, and two different 8-paths exist
 C The shortest m-path is 2 steps long D No path exists under any of the three rules

23 A 64×64 image with $L = 8$ has 790, 1023, 850, 656, 329, 245, 122 and 81 pixels at levels 0 to 7. After histogram equalization, what does the level $r = 2$ become? 3 MARKS

A 3 B 4
C 5 D 6

24 For that same image, how many distinct gray levels remain after histogram equalization? 3 MARKS

A 8 B 7
C 5 D 4

25 Why is the histogram after discrete equalization not perfectly uniform, even though the continuous proof says it should be? 3 MARKS

A Because the formula is wrong for digital images B Because rounding forces several input levels onto the same output level
C Because the image is too small D Because the cumulative distribution is not monotonic

26 x is 2×2 and h is 2×2 . What is the size of the block Toeplitz matrix used to compute their convolution? 3 MARKS

A 4×4 B 9×4
C 9×9 D 16×4

27 For the two matrices below, what is $y(1,1)$, the centre value of the 3×3 convolution result? 3 MARKS

$x = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $h = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

A 32 B 40
C 60 D 22

28 Which check can you always use to verify a 2D convolution answer? 3 MARKS

A The result must come out square B The sum of all values in y equals (sum of x) $\times$ (sum of h)
C y must contain no zeros D $y(0,0)$ is always zero

29 Why can the D_m distance between two fixed pixels change while D_4 and D_8 between them stay the same? 3 MARKS

A Because D_4 and D_8 also change, but more slowly B Because m -adjacency refuses a corner step whenever a common 4-neighbour is in V , so the path has to detour
C Because the coordinates of the pixels change D It cannot change — D_m depends only on the coordinates

30 A monitor has a γ of 2.5. What transformation must be applied to the image before it is displayed? 3 MARKS

- | | |
|--------------------------------|------------------------------------|
| A $s = c \cdot r^{2.5}$ | B $s = c \cdot r^{0.4}$ |
| C $s = 255 - r$ | D $s = c \cdot \log(1 + r)$ |

31 In histogram matching, a pixel has $sk = 3$ after equalization. The rounded table of $G(zq)$ for $q = 0$ to 7 is 0, 0, 0, 1, 2, 5, 6, 7. Which zq does it map to? 3 MARKS

- A** Z_3 **B** Z_4
C Z_5 **D** Z_6

32 Counting the centre pixel itself, how many pixels satisfy $D_4 \leq 2$ from a given pixel? 3 MARKS

- A** 9 **B** 13
C 25 **D** 5

33 The pixel value 167 is rebuilt using only bit planes 8 and 7. What value results? 3 MARKS

167 in binary = 1 0 1 0 0 1 1 1
plane weights = 128 64 32 16 8 4 2 1

- A** 160 **B** 128
C 167 **D** 64

34 Why does 4-connectivity usually find more objects in a binary image than 8-connectivity does? 3 MARKS

- A** Because it scans the image faster
- B** Because it refuses corner contacts, so pixels touching only at a corner are counted as separate objects
- C** Because it ignores the background
- D** Because it uses a larger neighbourhood

35 In a piecewise linear contrast stretch, what happens if $r_1 = r_2$, $s_1 = 0$ and $s_2 = L - 1$? 3 MARKS

- A** Nothing changes, because the function stays linear
- B** It becomes a thresholding function, producing a binary image
- C** It becomes the negative transformation
- D** It becomes the identity transformation

36 A kernel made of four 1s, one step away from the origin in each direction, has the frequency response $2\cos \omega_1 + 2\cos \omega_2$. What kind of filter is it? 3 MARKS

- A** A high-pass sharpening filter
- B** A low-pass smoothing filter, because the response peaks at $\omega = 0$
- C** A band-reject filter
- D** An all-pass filter

37 In the adaptive rule $g = C \cdot f$ applied when $m_{Sxy} \leq k_0 \sigma_G$ AND $k_1 \sigma_G \leq \sigma_{Sxy} \leq k_2 \sigma_G$, what does the lower limit $k_1 \sigma_G$ reject? 3 MARKS

- A** Bright regions of the image
- B** Completely flat patches, which contain nothing to reveal
- C** Regions that already have high contrast
- D** Pixels on the border of the image

- 38** Why does global histogram equalization fail to reveal small squares hidden inside larger dark squares? 3 MARKS
- A Because the small squares are brighter than the background B Because they are too close in gray level to the larger squares and too small in size to influence the global histogram
- C Because equalization only works on colour images D Because the histogram has been normalized
- 39** Which of these conversions throws information away and cannot be undone? 3 MARKS
- A RGB to HSV B RGB to YCbCr
- C RGB to grayscale D YCbCr back to RGB
- 40** For two object pixels the formula gives $D_4 = 2$, yet no 4-path exists between them inside the object. Why is that not a contradiction? 3 MARKS
- A The formula is being applied wrongly B The formula assumes every pixel in between may be used, while a real path must stay on pixels whose value is in V
- C Because D_4 applies only to colour images D Because the image is too small for the formula to hold

The answer key

The letter, then why. Read the explanation even where you got it right — that is where most of the revision value is.

- 01 **B** x and y are the spatial coordinates, and the amplitude of f at any pair of coordinates is called the intensity or gray level of the image at that point. When x , y and f are all whole numbers, the image is a digital image.
- 02 **C** An 8-bit pixel holds one byte, so it can take 256 values from 0 to 255. By convention 0 is black and 255 is white, and everything in between is a shade of gray.
- 03 **A** Sampling fixes how many points of the scene are measured, so it fixes $M \times N$, the size of the image. Quantization fixes how many brightness values each of those points is allowed to take, so it fixes L , the number of gray levels.
- 04 **C** $L = 2^{\lceil \log_2 k \rceil}$. So 1 bit gives 2 levels, 4 bits give 16, and 8 bits give 256. The intensity values are whole numbers in the interval $[0, L-1]$.
- 05 **B** $N_4(p)$ holds the pixels above, below, left and right of p — the ones sharing a full side. The four corner pixels form $N_{\leq D}(p)$, and the two sets joined together make $N_8(p)$, which has eight.
- 06 **B** $D_4(p, q) = |x-s| + |y-t|$. It is called city-block because only horizontal and vertical moves are allowed, like walking in a city laid out in a rectangular grid of streets. Pixels at equal D_4 distance form a diamond.
- 07 **A** For an 8-bit image that is $s = 255 - r$. It reverses the intensity levels, and it is suited to enhancing white or gray detail embedded in dark regions, especially when the black area is dominant in size — an X-ray, for example.
- 08 **B** There is no general theory of image enhancement — it is problem-oriented, and a method that suits X-ray images may be wrong for infrared ones. Restoration is the opposite: it is objective, because it is built on a mathematical model of the degradation.
- 09 **D** Ordered by falling energy per photon: γ rays, X-rays, ultraviolet, visible, infrared, microwaves, radio waves. Wavelength runs the other way, so γ rays are the shortest and radio waves the longest.
- 10 **B** The histogram is a plot of the number of occurrences of each gray level against the gray level value. Dividing every n_k by the total number of pixels gives the normalized histogram $p(r_k) = n_k/MN$, whose components sum to 1.
- 11 **C** $256 = 2^8$, so $k = 8$. Then $b = M \times N \times k = 512 \times 512 \times 8 = 2,097,152$ bits. Divide by 8 for bytes: 262,144 bytes, which is 256 KB.
- 12 **B** The differences are 4 and 3. $D_4 = 4 + 3 = 7$, $D_8 = \max(4, 3) = 4$, and the Euclidean distance is $\sqrt{25} = 5$. Notice the order that always holds: $D_8 \leq D_e \leq D_4$.
- 13 **C** The rule is $(M_1+N_1-1) \times (M_2+N_2-1)$, applied to each direction separately: $(3+2-1) \times (3+2-1) = 4 \times 4$. Add the sizes and subtract one — never multiply them.
- 14 **B** A fractional γ bulges the curve above the diagonal, mapping a narrow range of dark inputs into a wide range of outputs, so the picture brightens. γ greater than 1 does the opposite and is the cure for a washed-out image.
- 15 **B** $p(r_k) = n_k/MN$ turns counts into probabilities. That is what allows the gray levels to be treated as a random variable, which is the whole basis of histogram equalization.
- 16 **B** 1,048,576 pixels become 65,536, so 983,040 are lost — 93.75 %. Notice the side is divided by 4 but the pixel count by 16, because area goes as the square.
- 17 **C** The higher-order bits contain the majority of the visually significant data. Plane 8 alone still looks like the picture, while plane 1 looks close to random noise.
- 18 **B** Under 8-adjacency two pixels can be joined in more than one way at once, so 'the path' between them is not unique. m-adjacency allows a corner step only when there is no way round through a common side neighbour, which leaves exactly one path.

- 19 **B** A Fourier spectrum can span 0 to 10^6 or more, so on a 256-level display everything but the brightest spike looks black. The log squeezes a range of 0 to 1.5×10^6 down to about 0 to 6.2, and the detail appears. It expands dark values and compresses bright ones.
- 20 **B** The shape changes from (rows, cols) to (rows, cols, 3), but copying one plane three times cannot invent colour that was thrown away earlier. The conversion changes the container, not the content.
- 21 **B** Test condition (ii): $N_4(0,2) \cap N_4(1,1) = \{(0,1), (1,2)\}$. The value at (0,1) is 1, which is in V , so a way round already exists and the corner step is refused. Compare (1,1) and (2,2), whose common neighbours (1,2) and (2,1) are both 0 — those two are m-adjacent.
- 22 **B** No 4-path exists at all, because (1,1) and (2,2) meet only at a corner. Under 8-adjacency the shortest route is $(0,2) \rightarrow (1,1) \rightarrow (2,2)$, but $(0,2) \rightarrow (0,1) \rightarrow (1,1) \rightarrow (2,2)$ is equally valid, so there are two. m-adjacency leaves exactly one path, of length 3.
- 23 **C** $MN = 4096$. The cumulative count up to level 2 is $790 + 1023 + 850 = 2663$, so the cumulative probability is 0.65. Then $s = 7 \times 0.65 = 4.55$, which rounds to 5.
- 24 **C** The eight values of s_k come out as 1, 3, 5, 6, 6, 7, 7, 7. Levels 3 and 4 both land on 6, and levels 5, 6 and 7 all land on 7, so only five distinct output levels survive: 1, 3, 5, 6 and 7.
- 25 **B** In the continuous case $ps(s) = 1/(L-1)$ exactly, whatever the input density. In the discrete case the values of s_k must be rounded to whole numbers, which merges levels and leaves gaps, so a uniform histogram cannot be guaranteed.
- 26 **B** The result is 3×3 , so it has 9 elements and the matrix needs 9 rows. The other matrix is written as a column of its 4 elements, so the matrix needs 4 columns. Rows = number of output pixels, columns = number of kernel pixels.
- 27 **C** Four products land on (1,1): $x(0,0)h(1,1) + x(0,1)h(1,0) + x(1,0)h(0,1) + x(1,1)h(0,0) = 1 \times 8 + 2 \times 7 + 3 \times 6 + 4 \times 5 = 8 + 14 + 18 + 20 = 60$. The full answer is [5 16 12; 22 60 40; 21 52 32].
- 28 **B** Two free checks exist. The corner $y(0,0)$ is always just $x(0,0) \cdot h(0,0)$, and the grand total obeys $\Sigma y = (\Sigma x)(\Sigma h)$. For $x = [1 \ 2; 3 \ 4]$ and $h = [5 \ 6; 7 \ 8]$ that is $10 \times 26 = 260$, and the 3×3 answer does add up to 260.
- 29 **B** D_4 and D_8 are formulas on the coordinates alone, so nothing else can affect them. $D_{\text{sub}m}$ is the length of the shortest m-path, and m-adjacency looks at the values of the neighbouring pixels. In the standard example the answer moves between 2, 3 and 4 as p_1 and p_3 change from 0 to 1.
- 30 **B** Γ correction pre-distorts the image with the inverse power, $s = c r^{\sup>1/\gamma</sup>} = c r^{\sup>0</sup>}.4$. The two non-linearities then cancel, since $(r^{\sup>0</sup>}.4)^{\sup>2</sup>}.5 = r$. Without it the display crushes the mid-tones and the picture looks too dark.
- 31 **B** Look down the table for the value closest to 3. $G(z_4) = 2$ is one away, while $G(z_5) = 5$ is two away, so $q = 4$. This is why the inverse of G never has to be computed — you tabulate G and search it.
- 32 **B** The D_4 contours form a diamond: 1 pixel at distance 0, 4 at distance 1 and 8 at distance 2, giving 13. For D_8 the contours are squares, so $D_8 \leq 2$ covers a 5×5 block, which is 25 pixels.
- 33 **B** Plane n contributes $2^{\sup>n-1</sup>}$. Plane 8 holds a 1 and is worth 128; plane 7 holds a 0 and is worth nothing. Everything below is discarded, so the rebuilt value is $128 + 0 = 128$. Keeping only plane 8 can produce just two values, 0 and 128.
- 34 **B** 4-adjacency is the strictest rule and 8-adjacency the loosest, so a picture usually breaks into more objects under 4-connectivity. m-connectivity gives the same object count as 8-connectivity, because it only removes a corner link when a route round already exists — it changes the paths, never the objects.
- 35 **B** The middle segment turns vertical, so every input below the control point goes to 0 and every input above it goes to $L-1$. The other special case to remember is $r_1 = s_1$ and $r_2 = s_2$, which gives a straight line and changes nothing at all.

- 36 **B** Substituting $z = e^{j\omega}$ and using $e^{j\theta} + e^{-j\theta} = 2\cos \theta$ gives that response. It reaches its maximum of 4 at $\omega_1 = \omega_2 = 0$ and falls away as the frequency rises, so low frequencies — the smooth areas — are passed and high frequencies are suppressed.
- 37 **B** The first condition picks patches that are darker than the image as a whole. The second says the patch must have some variation but not much: the lower limit throws away perfectly uniform patches where there is nothing hidden, and the upper limit $k_2\sigma_G$ throws away patches that already have good contrast.
- 38 **B** A global transformation is built from the gray-level content of the entire image, and a few hundred pixels cannot move a histogram computed from a quarter of a million. Local histogram equalization computes the transformation inside a small window, maps only the centre pixel, and so brings the small areas out.
- 39 **C** HSV and YCbCr only rewrite the same information in another form, so the original comes back to within a gray level or two of rounding. Grayscale replaces three numbers per pixel by one, and the colour is gone for good — which is why turning a gray image back into three channels only produces three identical planes.
- 40 **B** D_4 and D_8 are the shortest 4-path and 8-path lengths only when the whole rectangle between the two pixels is available. Inside a binary image the path is restricted to pixels in V , so the true path can be longer than the formula says, or may not exist at all.